

Exercise 2.9.1 Let K be an inertial (non-rotating) system with coordinates (T, X, Y, Z) and line element

$$c^2 d\tau^2 = c^2 dT^2 - dX^2 - dY^2 - dZ^2 \quad (2.84)$$

Denote $X^0 = cT$, $X^1 = X$, $X^2 = Y$, $X^3 = Z$.

Let K' be a rotating system with coordinates (t, x, y, z) , implicitly defined by

$$\begin{aligned} T &= t \\ X &= x \cos \omega t - y \sin \omega t \\ Y &= x \sin \omega t + y \cos \omega t \\ Z &= z \end{aligned} \quad (2.85)$$

Show that the line element in terms of K' is

$$\begin{aligned} g_{\mu\nu} dx^\mu dx^\nu &= c^2 d\tau^2 \\ &= [c^2 - \omega^2(x^2 + y^2)] dt^2 + 2\omega y dx dt - 2\omega x dy dt - dx^2 - dy^2 - dz^2 \end{aligned} \quad (2.86)$$

Solution:

Denote $x^0 = ct$, $x^1 = x$, $x^2 = y$, $x^3 = z$; $\dot{x}^0 = c \dot{t}$, $\dot{x}^1 = \dot{x}$, $\dot{x}^2 = \dot{y}$, $\dot{x}^3 = \dot{z}$; and $\ddot{x}^1 = \ddot{x}$, $\ddot{x}^2 = \ddot{y}$, $\ddot{x}^3 = \ddot{z}$, where dot represents differentiation by τ .

$$dT = dt$$

$$dX = dx \cos \omega t - dy \sin \omega t - \omega (x \sin \omega t + y \cos \omega t) dt$$

$$dY = dx \sin \omega t + dy \cos \omega t + \omega (x \cos \omega t - y \sin \omega t) dt$$

$$dZ = dz$$

$$dT^2 = dt^2$$

$$\begin{aligned} dX^2 &= dx^2 \cos^2 \omega t + dy^2 \sin^2 \omega t + \omega^2 (x^2 \sin^2 \omega t + y^2 \cos^2 \omega t \\ &\quad + 2xy \sin \omega t \cos \omega t) dt^2 - 2 dx dy \sin \omega t \cos \omega t - 2\omega (x \sin \omega t \cos \omega t \\ &\quad + y \cos^2 \omega t) dx dt + 2\omega (x \sin^2 \omega t + y \sin \omega t \cos \omega t) dy dt \end{aligned}$$

$$\begin{aligned} dY^2 &= dx^2 \sin^2 \omega t + dy^2 \cos^2 \omega t + \omega^2 (x^2 \cos^2 \omega t + y^2 \sin^2 \omega t \\ &\quad - 2xy \sin \omega t \cos \omega t) dt^2 + 2 dx dy \sin \omega t \cos \omega t \\ &\quad + 2\omega (x \sin \omega t \cos \omega t - y \sin^2 \omega t) dx dt \\ &\quad + 2\omega (x \cos^2 \omega t - y \sin \omega t \cos \omega t) dy dt \end{aligned}$$

$$dZ^2 = dz^2$$

$$\begin{aligned}
c^2 d\tau^2 &\stackrel{2.84}{=} c^2 dT^2 - dX^2 - dY^2 - dZ^2 \\
&= c^2 dt^2 - \left[(dx^2 + dy^2) (\sin^2 \omega t + \cos^2 \omega t) \right. \\
&\quad + \omega^2 (x^2 + y^2) (\sin^2 \omega t + \cos^2 \omega t) dt^2 \\
&\quad + (-2\omega y dx dt + 2\omega x dy dt) (\sin^2 \omega t + \cos^2 \omega t) \\
&\quad \left. + dz^2 \right] \\
&= c^2 dt^2 - \left[(dx^2 + dy^2) + \omega^2 (x^2 + y^2) dt^2 - 2\omega y dx dt \right. \\
&\quad \left. + 2\omega x dy dt \right] - dz^2 \\
&= [c^2 - \omega^2(x^2 + y^2)] dt^2 + 2\omega y dx dt - 2\omega x dy dt \\
&\quad - dx^2 - dy^2 - dz^2 \quad \blacksquare
\end{aligned}$$